

Farshad Moradi

professor

基本信息

姓名: Farshad Moradi

所属院校: 奥胡斯大学

地区: Denmark

海外经历: 有

学科领域

集成电路设计与集成系统

生物医学工程

研究领域

低功耗电路与能量管理

存储器设计与架构

生物信号处理与模式识别

生物医学传感与检测技术

数字集成电路设计

医用生物材料与植入器械

脑机接口与神经工程

模拟与射频IC设计

低功耗设计与能耗优化

混合信号集成技术

个人简介

Farshad Moradi is a Professor in the Department of Electrical and Computer Engineering at Aarhus University. His research focuses on neurotechnology, mixed-signal integrated circuits, and neuromorphic systems. With a substantial body of work, he has been cited over 2245 times, holding an h-index of 24. His contributions span various high-impact articles in the field of biointegrated devices, low-power electronics, and neuromorphic computing. Moradi collaborates with leading researchers across Europe and beyond, contributing significantly to the advancement of neurotechnology and integrated circuit design.

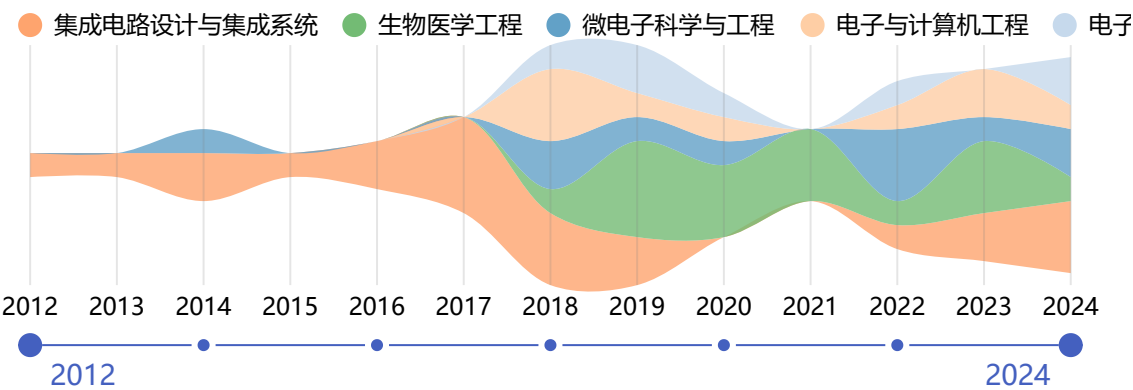
工作经历

教育经历

Universoty of Oslo - Ph.D.

2008-01-28 - 2011-10-09

近年发表的论文



Comparative analysis of THz signal emission from bilayer heterostructures: Wideband and high-frequency THz signal advantage of PtBi-based emitter

APPLIED PHYSICS LETTERS

Winkel, Tristan Joachim; Parvini, Tahereh Sadat; Stiewe, Finn-Frederik; Walowski, Jakob; Moradi, Farshad; Muenzenberg, Markus

被引用次数: 0 2024

Multimodal Temperature Sensing Using Electromechanical Impedance (EMI) for Structural Health Monitoring

IEEE SENSORS JOURNAL

Dahl, Mads Kofod; Hire, Jaamac Hassan; Zamani, Milad; Moradi, Farshad

被引用次数: 1 2024

Spin Hall Nano-Oscillator Empirical Electrical Model for Optimal On-Chip Detector Design

IEEE TRANSACTIONS ON ELECTRON DEVICES

Fiorelli, Rafaella; Rajabali, Mona; Mendez-Romero, Roberto; Kumar, Akash; Litvinenko, Artem; Serrano-Gotarredona, Teresa; Moradi, Farshad; Akerman, Johan; Linares-Barranco, Bernabe; Peralias, Eduardo

被引用次数: 0 2024

Temperature effect on a weighted vortex spin-torque nano-oscillator for neuromorphic computing

SCIENTIFIC REPORTS

Li, Ren; Rezaeiyan, Yasser; Bohnert, Tim; Schulman, Alejandro; Ferreira, Ricardo; Farkhani, Hooman; Moradi, Farshad

被引用次数: 1 2024

A System-Level Feasibility Study of a Lead-Free Ultrasonically Powered Light Delivery Implant for Optogenetics

ADVANCED INTELLIGENT SYSTEMS

Ivriq, Saeed Baghaee; Laursen, Kjeld; Jorgensen, Aske Moller; Mondal, Tanmay; Zamani, Milad; Rezaeiyan, Yasser; Corbett, Brian; Iversen, Bo Brummerstedt; Moradi, Farshad

被引用次数: 0

2024

A 65nm 3mA 0.14-m-Accuracy TDR Based Leak Detection SoC for District Heating Networks with I/C Calibration Technique

2024 IEEE CUSTOM INTEGRATED CIRCUITS CONFERENCE, CICC

Koolivand, Yarallah; Mosalmani, Alireza; Rezaeiyan, Yasser; Esmailbeygi, Hossein; Hatamzadeh, Elham; Zamani, Milad; Moradi, Farshad

被引用次数: 0

2024

A 69MHz-Bandwidth 40V/ μ s-Slew-Rate3nV/ $\sqrt{\text{Hz}}$ -Noise 4.5 μ V-Offset ChopperOperational Amplifier

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Koolivand, Yarallah; Rezaeiyan, Yasser; Zamani, Milad; Akbari, Meysam; Shoaee, Omid; Tang, Kea-Tiong; Moradi, Farshad

被引用次数: 0

2024

Vortex Spin Torque Nano Oscillator-based PUF and TRNG Design for Lightweight Security Solutions

8TH IEEE ELECTRON DEVICES TECHNOLOGY & MANUFACTURING CONFERENCE, EDTM 2024

Das, Kunal Kranti; Soni, Sandeep; Moradi, Farshad; Shreya, Sonal; Kumar, Brajesh; Kaushik

被引用次数: 1

2024

CMOS Front End for Interfacing Spin-Hall Nano-Oscillators for Neuromorphic Computing in the GHz Range

ELECTRONICS

Fiorelli, Rafaella; Peralias, Eduardo; Mendez-Romero, Roberto; Rajabali, Mona; Kumar, Akash; Zahedinejad, Mohammad; Akerman, Johan; Moradi, Farshad; Serrano-Gotarredona, Teresa; Linares-Barranco, Bernabe

被引用次数: 1

2023

A 4.5 μ W Miniaturized 3-Channel Wireless Intra-Cardiac Acquisition System

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Rezaeiyan, Yasser; Koolivand, Yarallah; Zamani, Milad; Shoaee, Omid; Akbari, Meysam; Moradi, Farshad; Tang, Kea-Tiong

被引用次数: 0

2023

NET-TEN: a silicon neuromorphic network for low-latency detection of seizures in local field potentials

JOURNAL OF NEURAL ENGINEERING

Ronchini, Margherita; Rezaeiyan, Yasser; Zamani, Milad; Panuccio, Gabriella; Moradi, Farshad

被引用次数: 3

2023

Granular vortex spin-torque nano oscillator for reservoir computing

SCIENTIFIC REPORTS

Shreya, S.; Jenkins, A. S.; Rezaeiyan, Y.; Li, R.; Bohnert, T.; Benetti, L.; Ferreira, R.; Moradi, F.; Farkhani, H.

被引用次数: 6

2023

Spin-orbit torque flash analog-to-digital converter

SCIENTIFIC REPORTS

Ghanatian, Hamdam; Benetti, Luana; Anacleto, Pedro; Bohnert, Tim; Farkhani, Hooman; Ferreira, Ricardo; Moradi, Farshad

被引用次数: 0

2023

An Ultrasonically-Powered System for 1.06 mm³ Implantable Optogenetics and Drug Delivery Dust

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Laursen, Kjeld; Zamani, Milad; Rezaeiyan, Yasser; Hosseini, Seyedsina; Mondal, Tanmay; Corbett, Brian; Ouagazzal, Abdel Mouttalib; Amalric, Marianne; Moradi, Farshad

被引用次数: 2

2023

Digital Hardware Implementation of ReSuMe Learning Algorithm for Spiking Neural Networks

2023 45TH ANNUAL INTERNATIONAL CONFERENCE OF THE IEEE ENGINEERING IN MEDICINE & BIOLOGY SOCIETY, EMBC

Khatiboun, Dario Fernandez; Rezaeiyan, Yasser; Ronchini, Margherita; Sadeghi, Maryam; Zamani, Milad; Moradi, Farshad

被引用次数: 1

2023

A low-noise high-linear wide dynamic-range MTJ-based magnetic field sensor

2023 IEEE NORDIC CIRCUITS AND SYSTEMS CONFERENCE, NORCAS

Rezaeiyan, Yasser; Madsen, Nikolaj Lykkeberg; Bohnert, Tim; Zamani, Milad; Shreya, Sonal; Paz, Elvira; Farkhani, Hooman; Ferreira, Ricardo; Moradi, Farshad

被引用次数: 0

2023

Enhancing spin-transfer torque in magnetic tunnel junction devices: Exploring the influence of capping layer materials and thickness on device characteristics

JOURNAL OF APPLIED PHYSICS

Parvini, Tahereh Sadat; Paz, Elvira; Bohnert, Tim; Schulman, Alejandro; Benetti, Luana; Oberbauer, Felix; Walowski, Jakob; Moradi, Farshad; Ferreira, Ricardo; Muenzenberg, Markus

被引用次数: 3

2023

A Hybrid Spin-CMOS Flash ADC based on Spin Hall Effect and Spin Transfer Torque

2022 IEEE 40TH INTERNATIONAL CONFERENCE ON COMPUTER DESIGN (ICCD 2022)

Ghanatian, Hamdam; Farkhani, Hooman; Moradi, Farshad

被引用次数: 0

2022

Energy Harvesting in Immersed Tunnel for Powering Wireless Sensor Nodes for Corrosion Monitoring

IEEE SENSORS JOURNAL

Hire, Jaamac Hassan; Agianniotis, Nikolaos; Kofoed, Brian Peter; Moradi, Farshad

Memory and Communication-in-Logic Using Vortex and Precessional Oscillations in a Magnetic Tunnel Junction

IEEE MAGNETICS LETTERS

Shreya, Sonal; Zamani, Milad; Rezaeiyan, Yaseer; Ghanatian, Hamdam; Bohnert, Tim; Jenkins, Alex S.; Ferreira, Ricardo; Farkhani, Hooman; Moradi, Farshad

被引用次数: 2

2022

Verilog-A-Based Analytical Modeling of Vortex Spin-Torque Nano Oscillator

IEEE TRANSACTIONS ON ELECTRON DEVICES

Shreya, Sonal; Rezaeiyan, Yasser; Jenkins, Alex; Bohnert, Tim; Farkhani, Hooman; Ferreira, Ricardo; Moradi, Farshad

被引用次数: 4

2022

A CMOS-based neuromorphic device for seizure detection from LFP signals

JOURNAL OF PHYSICS D-APPLIED PHYSICS

Ronchini, Margherita; Zamani, Milad; Hai Au Huynh; Rezaeiyan, Yasser; Panuccio, Gabriella; Farkhani, Hooman; Moradi, Farshad

被引用次数: 8

2022

A 3-Bit Flash Spin-Orbit Torque (SOT)-Analog-to-Digital Converter (ADC)

IEEE TRANSACTIONS ON ELECTRON DEVICES

Ghanatian, Hamdam; Farkhani, Hooman; Rezaeiyan, Yasser; Bohnert, Tim; Ferreira, Ricardo; Moradi, Farshad

被引用次数: 7

2022

STDP implementation using multi-state spin-orbit torque synapse

SEMICONDUCTOR SCIENCE AND TECHNOLOGY

Ghanatian, Hamdam; Ronchini, Margherita; Farkhani, Hooman; Moradi, Farshad

被引用次数: 5

2022

Optimum PZT Patch Size for Corrosion Detection in Reinforced Concrete Using the Electromechanical Impedance Technique

SENSORS

Hire, Jaamac Hassan; Hosseini, Seyedsina; Moradi, Farshad

被引用次数: 13

2021

Ultrasonic Backscatter Communication for Brain Implants: Mathematical Model, Simulation, and Measurement

2021 IEEE BIOMEDICAL CIRCUITS AND SYSTEMS CONFERENCE (IEEE BIOCAS 2021)

Christensen, Magnus N.; Zamani, Milad; Rashidi, Amin; Moradi, Farshad

被引用次数: 3

2021

Pain detection using batch normalized discriminant restricted Boltzmann machine layers

JOURNAL OF VISUAL COMMUNICATION AND IMAGE REPRESENTATION

Kharghanian, Reza; Peiravi, Ali; Moradi, Farshad; Iosifidis, Alexandros

被引用次数: 10

2021

S-MRUT: Sectored-Multiring Ultrasonic Transducer for Selective Powering of Brain Implants

IEEE TRANSACTIONS ON ULTRASONICS FERROELECTRICS AND FREQUENCY CONTROL

Hosseini, Seyed sina; Laursen, Kjeld; Rashidi, Amin; Mondal, Tanmay; Corbett, Brian; Moradi, Farshad

被引用次数: 9

2021

Biointegrated and Wirelessly Powered Implantable Brain Devices: A Review

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Das, Rupam; Moradi, Farshad; Heidari, Hadi

被引用次数: 109

2020

Ultrasonically Powered Compact Implantable Dust for Optogenetics

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Laursen, Kjeld; Rashidi, Amin; Hosseini, Seyed sina; Mondal, Tanmay; Corbett, Brian; Moradi, Farshad

被引用次数: 19

2020

Electronic Contact Lens: A Platform for Wireless Health Monitoring Applications

ADVANCED INTELLIGENT SYSTEMS

Yuan, Mengyao; Das, Rupam; Ghannam, Rami; Wang, Yinhao; Reboud, Julien; Fromme, Roland; Moradi, Farshad; Heidari, Hadi

被引用次数: 52

2020

LAO-NCS: Laser Assisted Spin Torque Nano Oscillator-Based Neuromorphic Computing System

FRONTIERS IN NEUROSCIENCE

Farkhani, Hooman; Bohnert, Tim; Tarequzzaman, Mohammad; Costa, Jose Diogo; Jenkins, Alex; Ferreira, Ricardo; Madsen, Jens Kargaard; Moradi, Farshad

被引用次数: 17

2020

A low-power, low-noise, high-accurate epileptic-seizure detection system for wearable applications

MICROELECTRONICS JOURNAL

Tohidi, Mohammad; Madsen, Jens Kargaard; Moradi, Farshad

被引用次数: 8

2019

An Implantable Ultrasonically Powered System for Optogenetic Stimulation with Power-Efficient Active Rectifier and Charge-Reuse Capability

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Rashidi, Amin; Laursen, Kjeld; Hosseini, Seyed sina; Huynh, Hai-Au; Moradi, Farshad

被引用次数: 19

2019

Low-Power High-Input-Impedance EEG Signal Acquisition SoC With Fully Integrated IA and Signal-Specific ADC for Wearable Applications

IEEE TRANSACTIONS ON BIOMEDICAL CIRCUITS AND SYSTEMS

Tohidi, Mohammad; Madsen, Jens Kargaard; Moradi, Farshad

被引用次数: 32	2019
An adaptive-resolution signal-specific ADC for sensor-interface applications	
ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING	
Nasserian, Mahshid; Peiravi, Ali; Moradi, Farshad	
被引用次数: 13	2019
A charge sharing-based switching scheme for SAR ADCs	
INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS	
Akbari, Meysam; Hashemipour, Omid; Nazari, Masoud; Moradi, Farshad	
被引用次数: 14	2019
LAS-NCS: A Laser-Assisted Spintronic Neuromorphic Computing System	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Farkhani, Hooman; Prejbeanu, Ioan Lucian; Moradi, Farshad	
被引用次数: 9	2019
Highly stable, low power FinFET SRAM cells with exploiting dynamic back-gate biasing	
INTEGRATION-THE VLSI JOURNAL	
Bagheriye, Leila; Toofan, Siroos; Saeidi, Roghayeh; Moradi, Farshad	
被引用次数: 7	2019
A Novel Nondestructive Bit-Line Discharging Scheme for Deep Submicrometer STT-RAMs	
IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING	
Zeinali, Behzad; Madsen, Jens Kargaard; Raghavan, Praveen; Moradi, Farshad	
被引用次数: 5	2019
Progressive Scaled STT-RAM for Approximate Computing in Multimedia Applications	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Zeinali, Behzad; Karsinos, Dimitrios; Moradi, Farshad	
被引用次数: 18	2018
A Low-Power High-Speed Spintronics-Based Neuromorphic Computing System Using Real-Time Tracking Method	
IEEE JOURNAL ON EMERGING AND SELECTED TOPICS IN CIRCUITS AND SYSTEMS	
Farkhani, Hooman; Tohidi, Mohammad; Farkhani, Sadaf; Madsen, Jens Kargaard; Moradi, Farshad	
被引用次数: 24	2018
A Reduced Store/Restore Energy MRAM-Based SRAM Cell for a Non-Volatile Dynamically Reconfigurable FPGA	
IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS	
Bagheriye, Leila; Toofan, Siroos; Saeidi, Roghayeh; Zeinali, Behzad; Moradi, Farshad	
被引用次数: 10	2018
Input Offset Estimation of CMOS Integrated Circuits in Weak Inversion	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Akbari, Meysam; Hashemipour, Omid; Moradi, Farshad	

被引用次数: 16	2018
Offset-Compensated High-Speed Sense Amplifier for STT-MRAMs	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Bagheriye, Leila; Toofan, Siroos; Saeidi, Roghayeh; Moradi, Farshad	
被引用次数: 14	2018
A fully-integrated 16-channel EEG readout front-end for neural recording applications	
AEU-INTERNATIONAL JOURNAL OF ELECTRONICS AND COMMUNICATIONS	
Nasserian, Mahshid; Peiravi, Ali; Moradi, Farshad	
被引用次数: 19	2018
An energy-efficient DAC switching algorithm based on charge recycling method for SAR ADCs	
MICROELECTRONICS JOURNAL	
Akbari, Meysam; Hashemipour, Omid; Khateb, Fabian; Moradi, Farshad	
被引用次数: 10	2018
A Domino Circuit Technique for Noise-Immune High Fan-In Gates	
JOURNAL OF CIRCUITS SYSTEMS AND COMPUTERS	
Asyaei, Mohammad; Moradi, Farshad	
被引用次数: 16	2018
STT-RAM Energy Reduction Using Self-Referenced Differential Write Termination Technique	
IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS	
Farkhani, Hooman; Tohidi, Mohammad; Peiravi, Ali; Madsen, Jens Kargaard; Moradi, Farshad	
被引用次数: 29	2017
Quasi-Schottky-Barrier UTBB SOI MOSFET for Low-Power Robust SRAMs	
IEEE TRANSACTIONS ON ELECTRON DEVICES	
Ghanatian, Hamdam; Hosseini, Seyed Ebrahim; Zeinali, Behzad; Moradi, Farshad	
被引用次数: 2	2017
Design and Analysis of an Ultra-Low-Power Second-Order Asynchronous Delta-Sigma Modulator	
CIRCUITS SYSTEMS AND SIGNAL PROCESSING	
Akbari, Meysam; Hashemipour, Omid; Moradi, Farshad	
被引用次数: 5	2017
Low-leakage sub-threshold 9T-SRAM cell in 14-nm FinFET technology	
INTERNATIONAL JOURNAL OF CIRCUIT THEORY AND APPLICATIONS	
Zeinali, Behzad; Madsen, Jens Kargaard; Raghavan, Praveen; Moradi, Farshad	
被引用次数: 17	2017
Low-Energy Write Operation for 1T-1MTJ STT-RAM Bitcells With Negative Bitline Technique	

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Farkhani, Hooman; Peiravi, Ali; Moradi, Farshad

被引用次数: 11

2016

Graphene-Based STT-RAM Cell With Improved Switching for Scaled Technology Nodes

IEEE TRANSACTIONS ON MAGNETICS

Varghani, Ali; Peiravi, Ali; Moradi, Farshad

被引用次数: 3

2016

A low-power fast tag comparator by modifying charging scheme of wide fan-in dynamic OR gates

INTEGRATION-THE VLSI JOURNAL

Nasserian, Mahshid; Kafi-Kangi, Mohammad; Maymandi-Nejad, Mohammad; Moradi, Farshad

被引用次数: 21

2016

A new write assist technique for SRAM design in 65 nm CMOS technology

INTEGRATION-THE VLSI JOURNAL

Farkhani, Hooman; Peiravi, Ali; Moradi, Farshad

被引用次数: 6

2015

Multi-level wordline driver for robust SRAM design in nano-scale CMOS technology

MICROELECTRONICS JOURNAL

Moradi, Farshad; Panagopoulos, Georgios; Karakostas, Georgios; Farkhani, Hooman; Wisland, Dag T.; Madsen, Jens K.; Mahmoodi, Hamid; Roy, Kaushik

被引用次数: 12

2014

A new asymmetric 6T SRAM cell with a write assist technique in 65 nm CMOS technology

MICROELECTRONICS JOURNAL

Farkhani, Hooman; Peiravi, Ali; Moradi, Farshad

被引用次数: 24

2014

Domino logic designs for high-performance and leakage-tolerant applications

INTEGRATION-THE VLSI JOURNAL

Moradi, Farshad; Tuan Vu Cao; Vatajelu, Elena I.; Peiravi, Ali; Mahmoodi, Hamid; Wisland, Dag T.

被引用次数: 46

2013

Low phase-noise VCO utilizing modified symmetric load and partial positive feedback for FDSM

ANALOG INTEGRATED CIRCUITS AND SIGNAL PROCESSING

Tuan-Vu Cao; Wisland, Dag T.; Lande, Tor Sverre; Moradi, Farshad

被引用次数: 1

2012

Asymmetrically Doped FinFETs for Low-Power Robust SRAMs

IEEE TRANSACTIONS ON ELECTRON DEVICES

Moradi, Farshad; Gupta, Sumeet Kumar; Panagopoulos, Georgios; Wisland, Dag T.; Mahmoodi, Hamid; Roy, Kaushik

Residual effects of nitrogen fertilization on foliar chlorophyll and yield of the succeeding rice crop

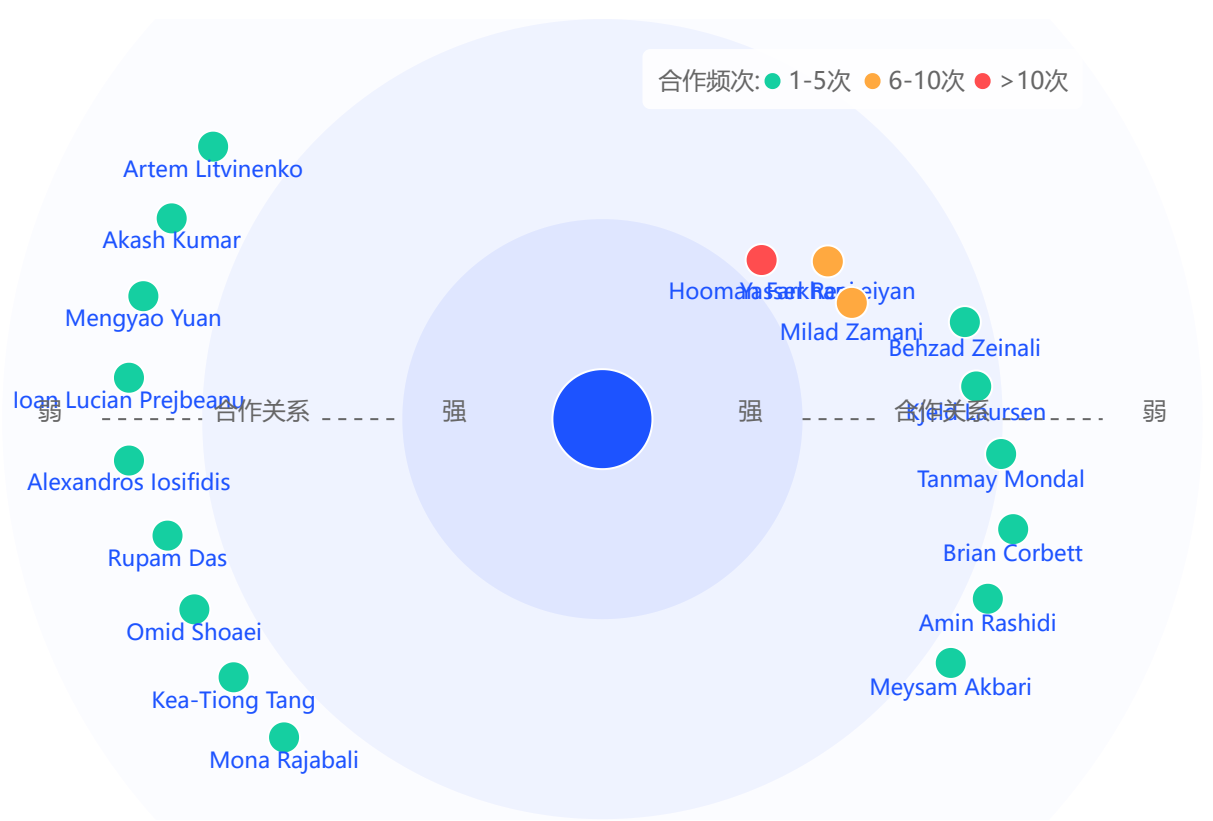
PHILIPPINE AGRICULTURAL SCIENTIST

Moradi, F; Alejar, AA; Peng, S; Visperas, RM

被引用次数：0

2002

学术合作



Hooman Farkhani	合作次数：11
奥胡斯大学	
Yasser Rezaeiyan	合作次数：9
奥胡斯大学	
Milad Zamani	合作次数：9
奥胡斯大学	
Behzad Zeinali	合作次数：5
奥胡斯大学	
Kjeld Laursen	合作次数：5
奥胡斯大学	

还有 13 个合作者